Lab 1 Requirements Table – PES2UG24CS036

Micro-Lending & Peer Credit Risk Assessor

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall compute borrower credit risk ratings based on debt-to-income ratio, repayment history, and employment stability indicators.	High	For a complete financial profile, the system assigns exactly one risk tier: Low, Medium, or High; if any required indicator is missing, no credit approval is generated.	Core borrower risk-assessment function.
FR-002	Functional	The system shall allow a borrower to submit and update the financial information required for credit-risk assessment.	High	A borrower can save a complete profile containing debt-to-income information, repayment history, and employment stability; incomplete submission is rejected with the missing fields identified.	Provides the inputs required by FR-001.
FR-003	Functional	The system shall generate a structured repayment schedule for an approved loan request, including installment amounts and due dates.	High	For every approved loan, the generated schedule contains all installments, an amount for each installment, and a due date for each installment, with no missing entries.	Required to support organized loan repayment.
FR-004	Functional	The system shall record loan repayments and update the outstanding loan balance after a successful repayment transaction.	High	After a successful repayment, the transaction is recorded and the outstanding balance decreases by the repayment amount exactly once.	Supports accurate repayment tracking and balance management.
FR-005	Functional	The system shall automatically trigger a default warning when a scheduled repayment becomes overdue according to the loan repayment schedule.	High	For a deliberately overdue test repayment, a default warning is generated once for that overdue event and is associated with the correct loan.	Implements the automated default-warning behavior in the scenario.
NFR-001	Nonfunctional	The system shall execute loan repayment transactions and EMI distribution inside database ACID transactions to prevent double-spending.	High	Benchmarking and concurrency tests confirm that repayment and EMI distribution remain atomic and consistent under simulated peak load, with no duplicated allocation.	Protects financial correctness and transaction integrity.
NFR-002	Nonfunctional	The system shall restrict access to borrower financial and repayment information to authenticated users authorized for the associated loan.	High	A test suite confirms that an unauthorized user cannot retrieve borrower financial or repayment data, while authorized users can access the data required for their role.	Protects sensitive financial information from unauthorized access.

Use-Case Flow Specification

Assess Borrower Credit Risk

Primary Actor	Borrower
Secondary Actor	Peer Lender
Purpose	Compute a borrower risk tier from the required financial indicators.

Precondition

- The borrower is authenticated and has permission to submit or update a financial profile.
- The financial profile contains the inputs required for assessment: debt-to-income ratio, repayment history, and employment stability indicators.

Postcondition

- The system stores the completed risk assessment and assigns the borrower exactly one risk tier: Low, Medium, or High.
- If the profile is incomplete, no credit approval is generated.

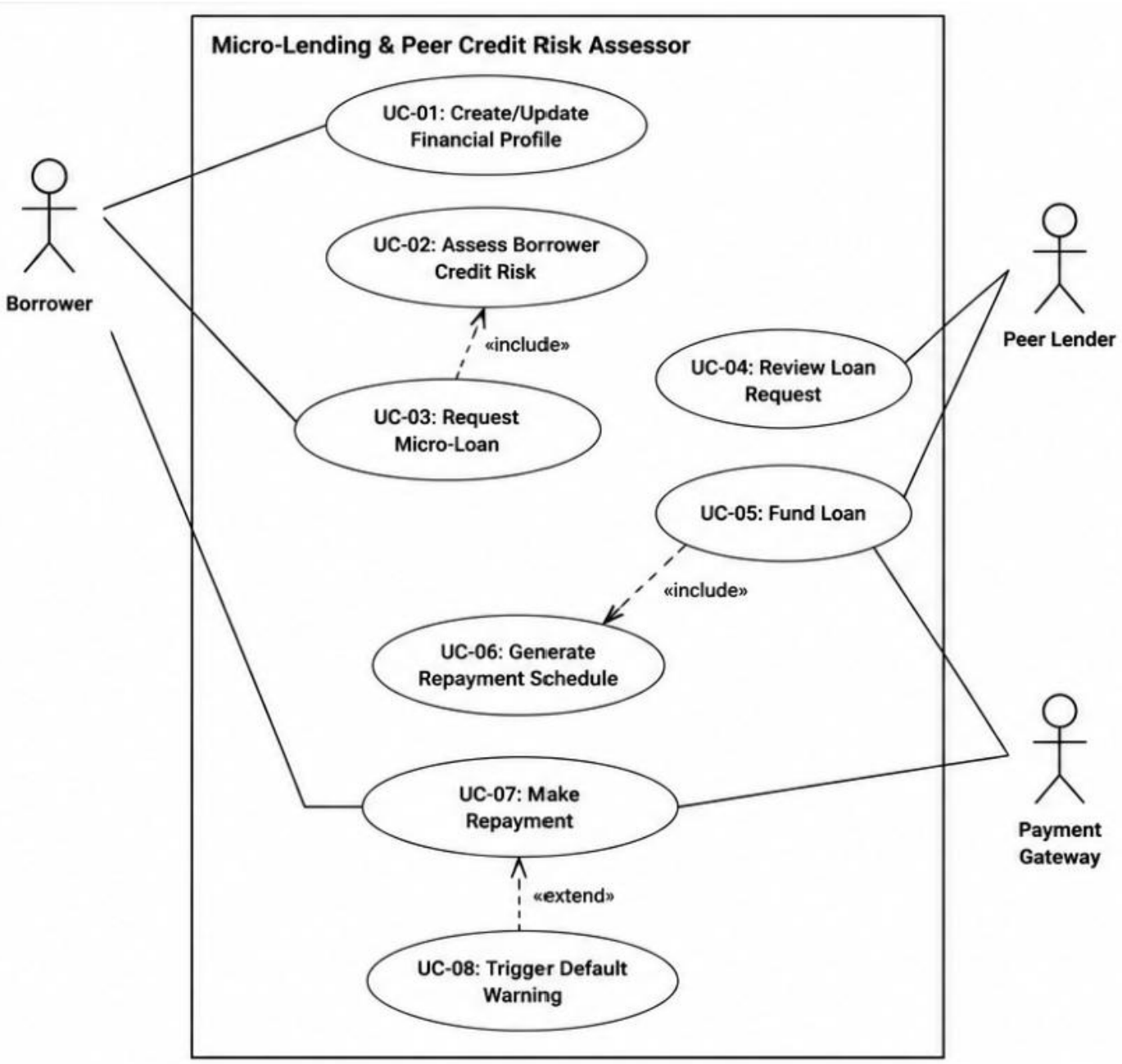
Main Success Scenario

1. Borrower opens the credit-risk assessment function and submits the required financial profile.
2. System validates that the debt-to-income ratio, repayment history, and employment stability indicators are present and valid.
3. System computes the borrower credit risk rating using the defined heuristic rules.
4. System assigns a risk tier (Low, Medium, or High).
5. System stores the assessment result for the associated borrower profile and loan decision.
6. System displays the assigned risk tier to the authorized user.
7. Use case ends successfully.

Alternate Flow - Incomplete Financial Profile

1. System detects that one or more required financial indicators are missing.
2. System identifies the missing information and does not generate a credit approval.
3. Borrower is prompted to complete the missing information; the use case may resume from Step 1.

Use-Case Diagram


Micro-Lending & Peer Credit Risk Assessor
UC-01: Create/Update Financial Profile
UC-02: Assess Borrower Credit Risk
«include»
UC-03: Request Micro-Loan
UC-04: Review Loan Request
UC-05: Fund Loan
«include»
UC-06: Generate Repayment Schedule
UC-07: Make Repayment
«extend»
UC-08: Trigger Default Warning
Borrower
Peer Lender
Payment Gateway